

Satellite Structural Inspection Interferometric Synthetic Aperture Radar (InSAR) Based

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METHODOLOGY

1. Satellite SAR System with Resolution and Revisit Time

To select a suitable SAR system, technical specifications such as spatial resolution, frequency band, polarization, and revisit time were considered. Sentinel-1, operating at C-band (5.405 GHz), was selected due to:

Parameter	Sentinel-1 C-SAR (IW Mode)
Frequency Band	C-band (~5.405 GHz)
Revisit Time	12 days (single satellite)
Spatial Resolution	5 m (range) × 20 m (azimuth)
Coverage Mode	Interferometric Wide Swath (IW)

2. Retrieving Multi-Temporal SAR Imagery Over the Specified Period

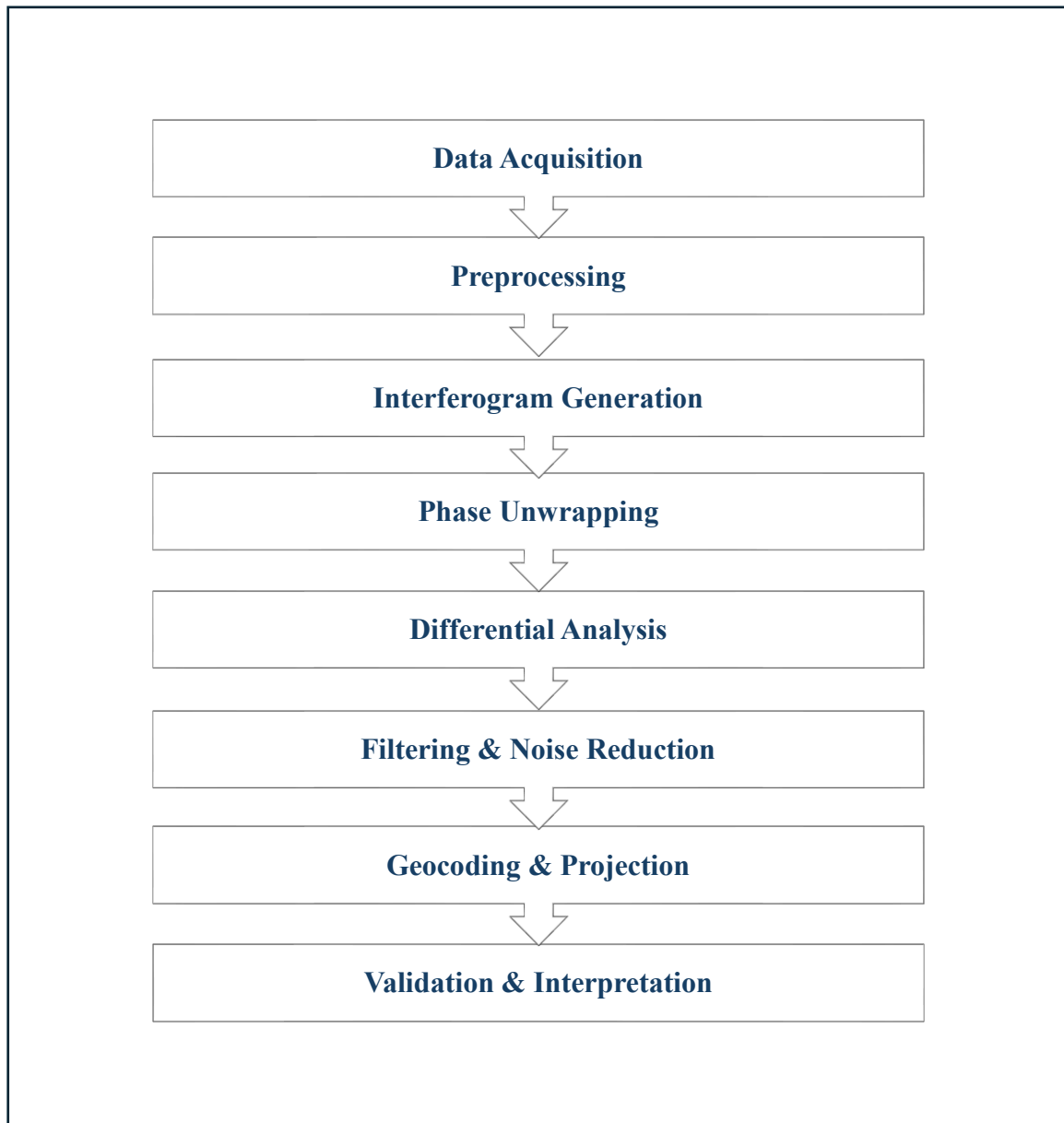
Multi-temporal **Sentinel-1 SLC IW mode** data were downloaded based on:

- **Area of Interest (AOI):** A polygon covering the bridge and surrounding terrain.
- **Date range:** Multiple acquisitions were collected spanning several months to capture temporal variations.
- **Image acquisition conditions:** All images were selected with the same relative orbit, pass direction (ascending or descending), and polarization (typically VV/VH) to ensure consistency for interferometric analysis.

S1A_IW_SLC__1SDV_20250111T130240_20250111T130310_057397_0710AB_813D
S1A_IW_SLC__1SDV_20250123T130239_20250123T130309_057572_07179E_5431
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S1A_IW_SLC__1SDV_20250216T130238_20250216T130308_057922_072592_C312
S1A_IW_SLC__1SDV_20250228T130238_20250228T130307_058097_072CB4_7A14
S1A_IW_SLC__1SDV_20250312T130238_20250312T130308_058272_0733CA_217F
S1A_IW_SLC__1SDV_20250324T130238_20250324T130308_058447_073AB5_B445
S1A_IW_SLC__1SDV_20250405T130238_20250405T130308_058622_0741CE_2B52
S1A_IW_SLC__1SDV_20250417T130238_20250417T130308_058797_0748F4_3FFD
S1A_IW_SLC__1SDV_20250429T130239_20250429T130309_058972_07501E_08D8
S1A_IW_SLC__1SDV_20250511T130238_20250511T130308_059147_0756BA_DE13
S1A_IW_SLC__1SDV_20250523T130238_20250523T130308_059322_075CCB_14B2
S1A_IW_SLC__1SDV_20250604T130237_20250604T130307_059497_0762D9_3AB5
S1A_IW_SLC__1SDV_20250616T130237_20250616T130306_059672_0768D1_F57D
S1A_IW_SLC__1SDV_20250628T130236_20250628T130306_059847_076EE3_6BFD

3. Differential SAR Interferometry (D-InSAR) for Displacement Measurement

D-InSAR analysis was performed to measure millimetre-scale deformation over time:



VISUALIZATIONS

1. Defining specific bridge location & parameters for satellite SAR based inspection



Project & Bridge Details

State – Rajasthan

Bridge No. – 899/1

Bridge Chainage – 907+700

Project - Chittorgarh to Kota

Zone – RJ 06

Structure Type – VUP

Sar Parameters

Sensor – Sentinel-1 (C-Band)

Beam Mode – IW (Interferometric Wide Swath)

Path – 34

Frame – 519

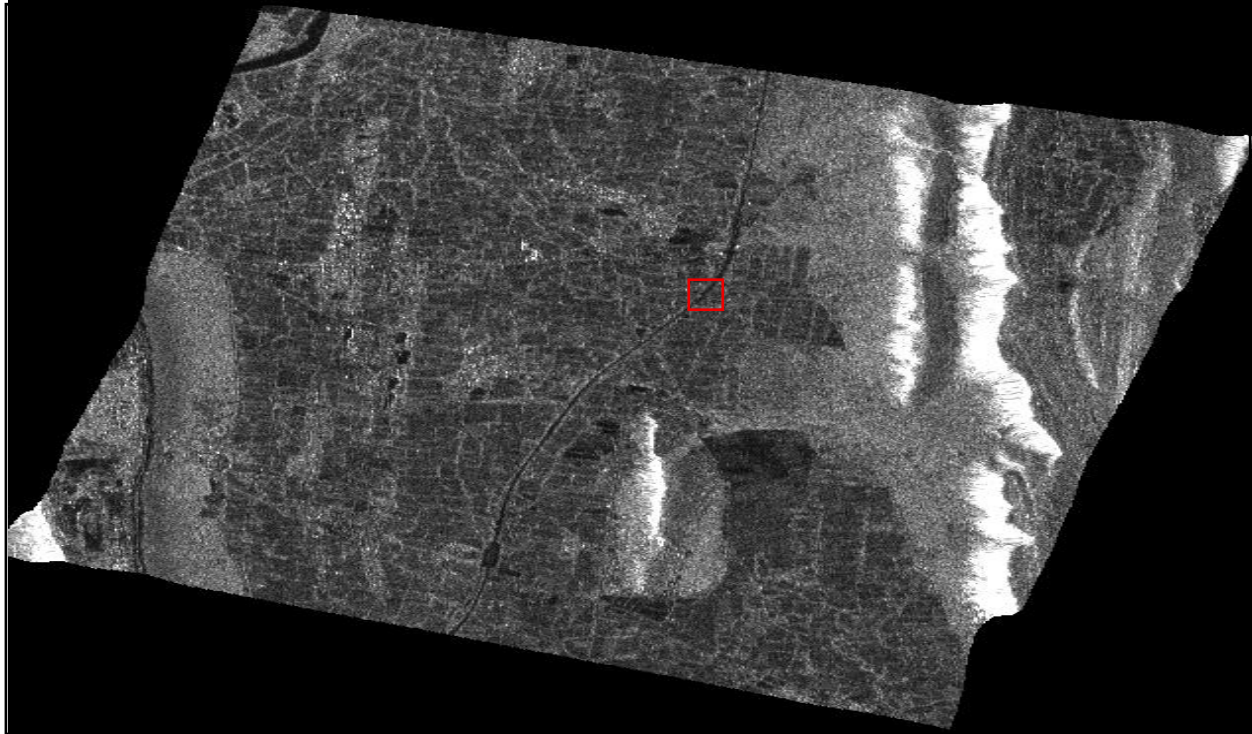
Flight Direction - DESCENDING

Polarization – VV+VH

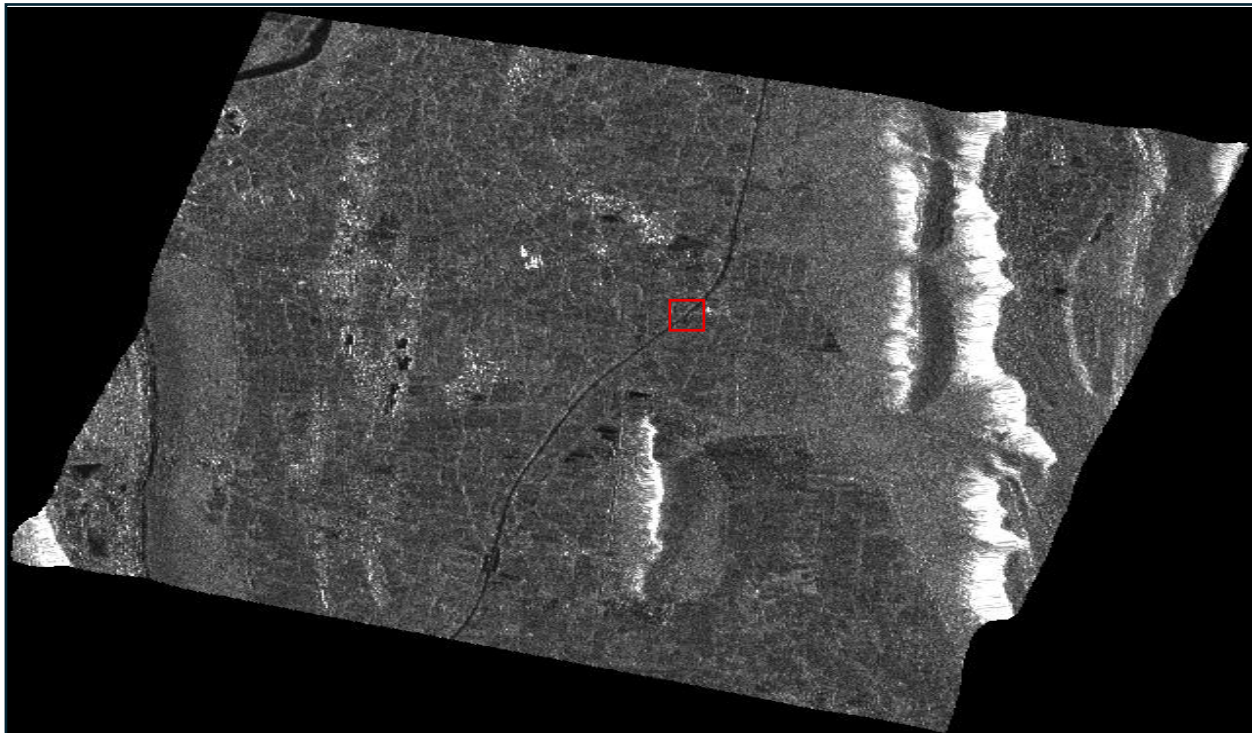
Absolute Orbit – 54706

2. Calibration, georeferencing, orthorectification of SAR images (VH/VV)

- Orthorectification of SAR image VH Polarization



- Orthorectification of SAR image VV Polarization



3. Results of SAR based bridge inspection, including visualizations

Bridge Components

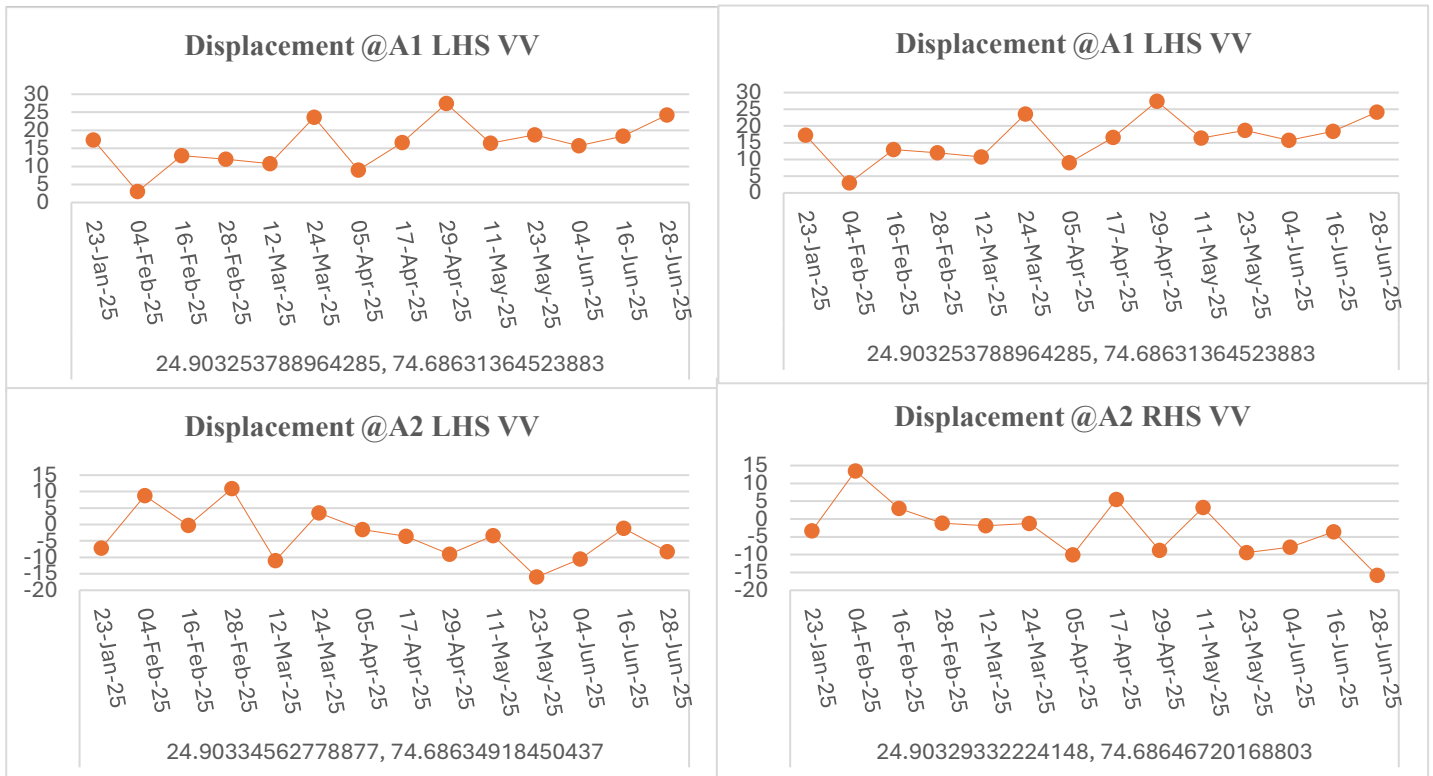
VUP – CH-907+700

Length – 23.43M

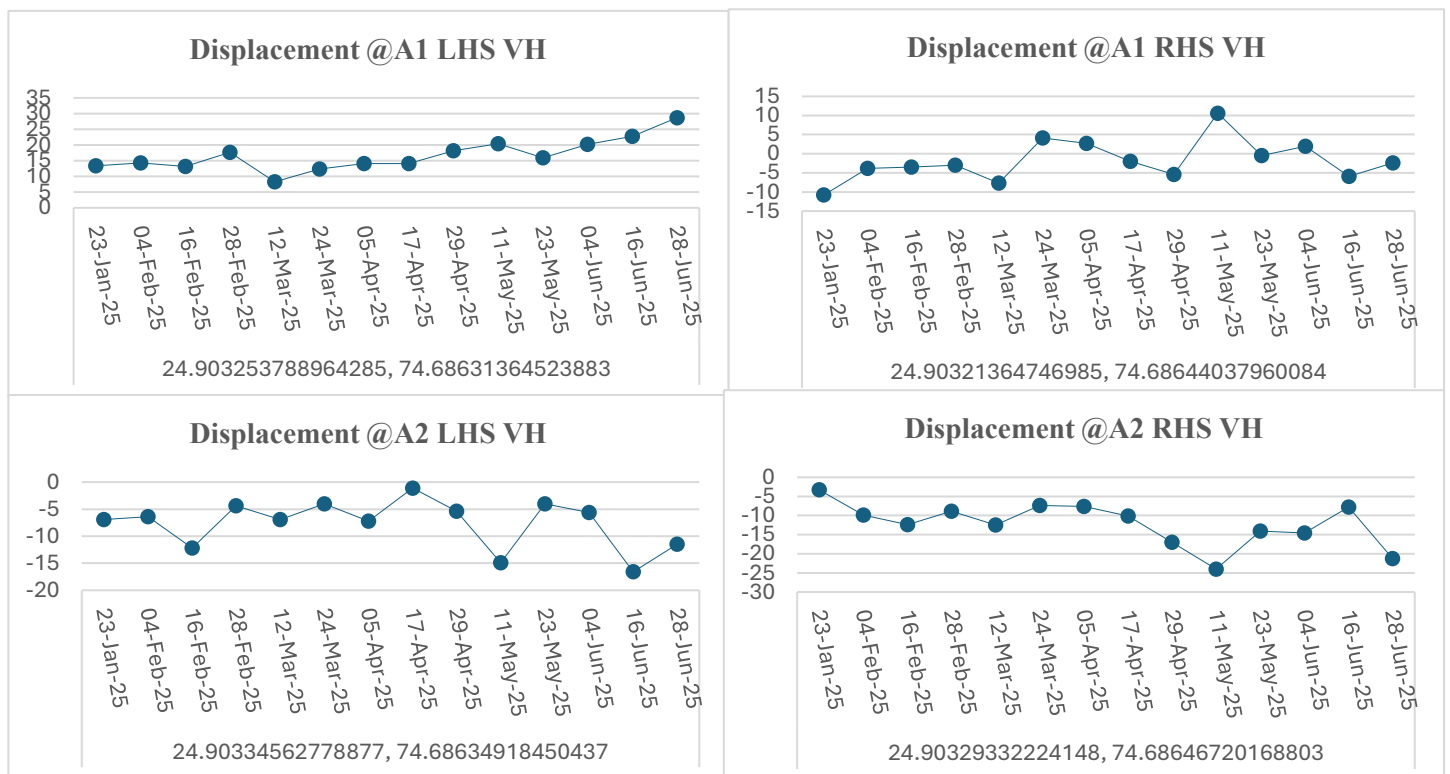


Scan Duration		From : 23 Jan 2025	To : 28 June 2025
Components & Abbreviation			
A1	Abutment 1		
A2	Abutment 2		
LHS	Left Hand Side		
RHS	Right Hand Side		

3.1 VV Displacement From 23 Jan 2025 To 28 June 2025



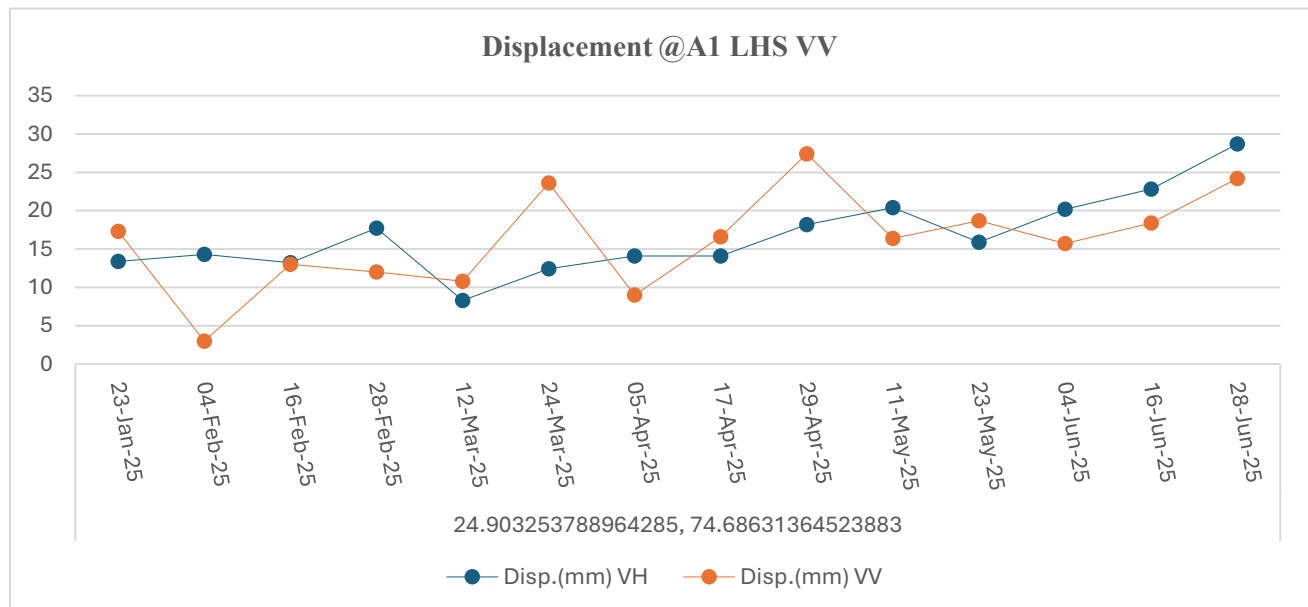
3.2 VH Displacement From 23 Jan 2025 To 28 June 2025



3.3 SAR Based Analysis at Abutment 1 LHS:



A1 LHS (24.903253788964285, 74.68631364523883)



Key Observations:

Displacement Magnitude Patterns:

- **Vertical Displacement (VV)** Vertical displacement at A1 LHS shows a strongly positive and increasing trend, indicating progressive uplift over the monitoring period.
- **Horizontal Displacement (VH)** Horizontal displacement also remains predominantly positive, ranging from approximately +8 mm to +28 mm.

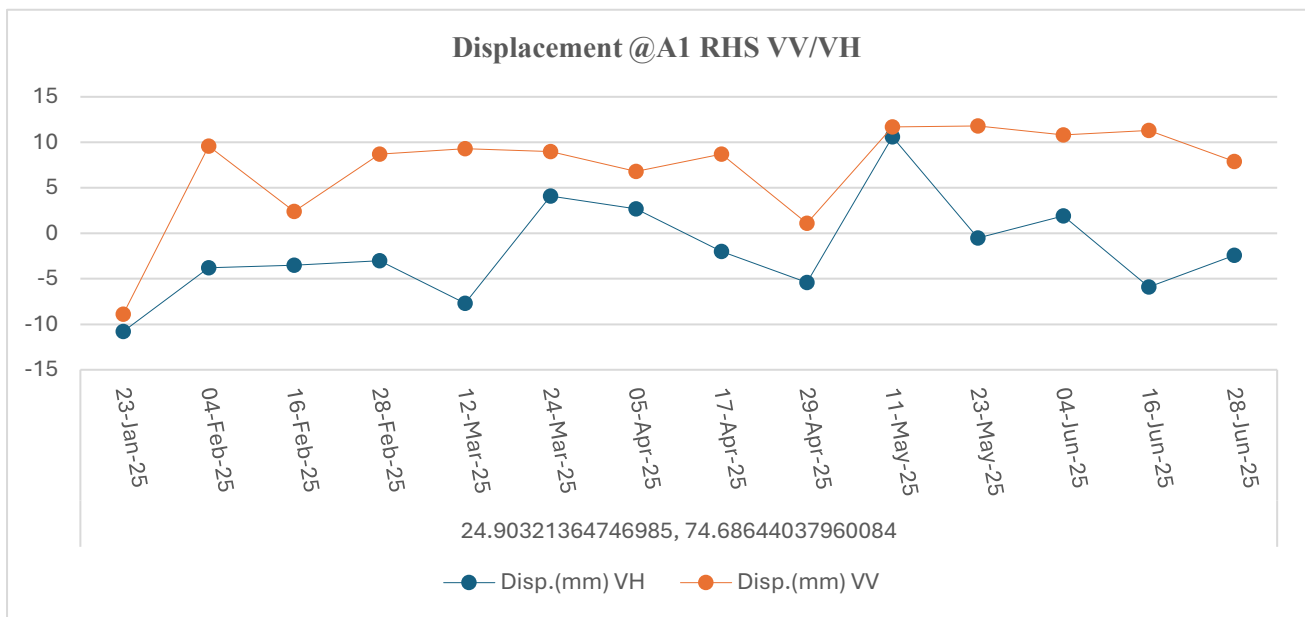
Critical Events & Anomalies:

- **February 2025:** VV drops sharply to +3 mm, while VH remains elevated (+14 mm), representing a temporary vertical relaxation without lateral reversal.
- **May 2025:** VH reaches +20 mm, while VV decreases slightly to +16 mm, indicating continued lateral accumulation following vertical peak.
- **June 2025:** VH rises steadily to +28 mm, while VV increases to +24 mm, representing the highest displacement values of the entire monitoring period, indicating active and continuing movement.

3.4 SAR Based Analysis at Abutment 1 RHS:



A1 RHS (24.90321364746985, 74.68644037960084)



Key Observations:

Displacement Magnitude Patterns:

- **Horizontal displacement (VH)** Horizontal displacement remains mostly negative, ranging from approximately -11 mm to +10 mm.
- **Vertical displacement (VV)** Vertical displacement at A1 RHS is predominantly positive, indicating sustained uplift behaviour throughout the monitoring period.

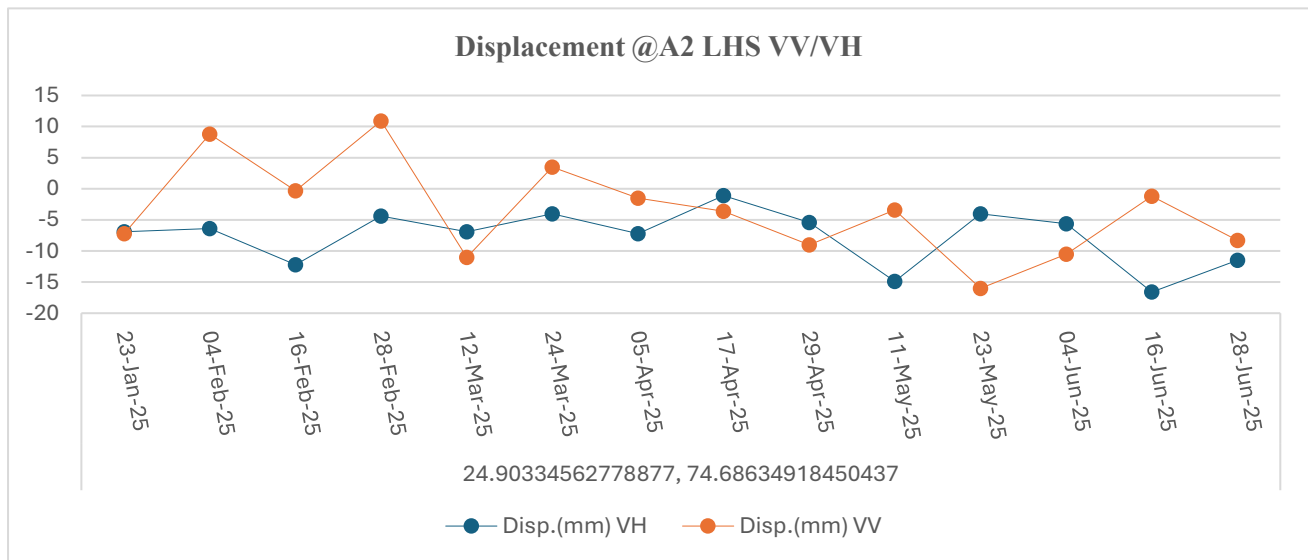
Critical Events & Anomalies:

- **January 2025:** VH begins near -11 mm, while VV is around -9 mm, indicating initial downward and lateral movement, likely representing the baseline condition.
- **April 2025:** Both VH and VV decrease slightly (VH +2 mm, VV +6 mm), representing a brief stabilisation phase
- **June 2025:** VH becomes negative again (-6 mm to -3 mm), while VV remains positive (+8-11 mm), confirming continued uplift without lateral stabilisation.

3.5 SAR Based Analysis at Abutment 2 LHS:



A2 LHS (24.90334562778877, 74.68634918450437)



Key Observations:

Displacement Magnitude Patterns:

- **Horizontal displacement (VH)** Horizontal displacement remains predominantly negative, varying roughly between -4 mm and -17 mm.
- **Vertical Displacement (VV)** at A2 LHS shows high variability with repeated sign reversals, ranging approximately from $+11$ mm uplift to -16 mm downward settlement.

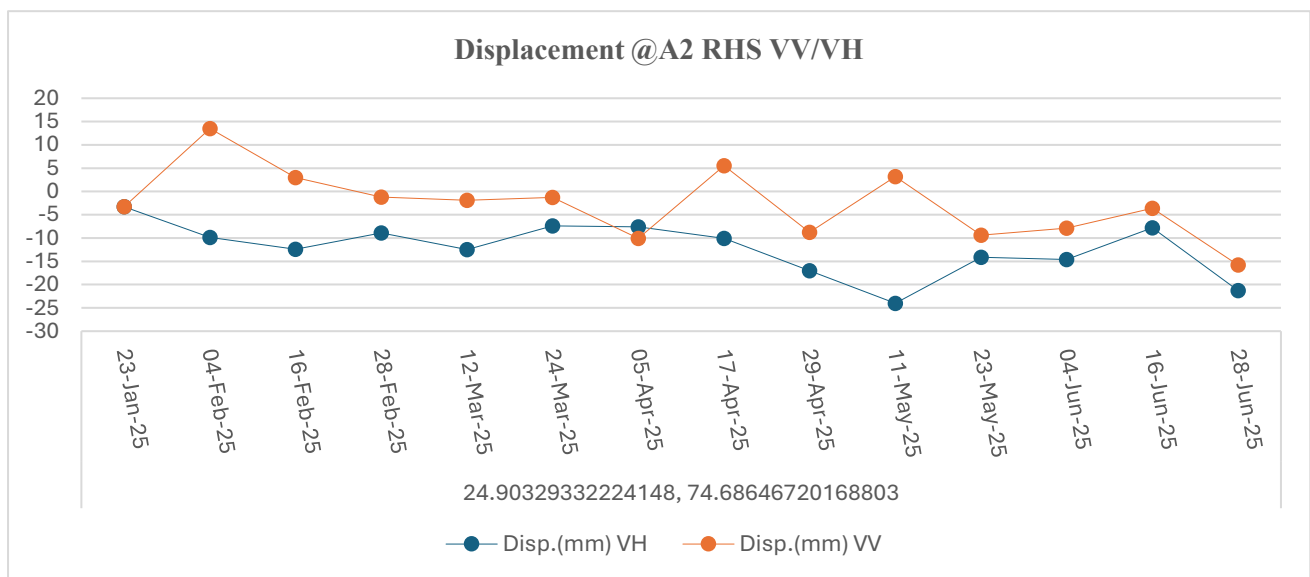
Critical Events & Anomalies:

- **February 2025:** VV rises sharply to $+8$ – 9 mm uplift, while VH remains negative (-7 mm), highlighting a strong vertical rebound without horizontal correction.
- **June 2025:** VH remains significantly negative (-17 to -12 mm), while VV oscillates between -1 mm and -9 mm, indicating persistent lateral movement with incomplete vertical recovery.

3.6 SAR Based Analysis at Abutment 2 RHS:



A2 RHS (24.90329332224148, 74.68646720168803)



Key Observations:

Displacement Magnitude Patterns:

- **Vertical (VV) displacement** Vertical displacement at A2 RHS shows alternating uplift and settlement with an overall downward tendency toward the later stages.
- **Horizontal (VH) displacement** Horizontal displacement is consistently negative throughout the monitoring period, varying roughly between -8 mm and -24 mm.

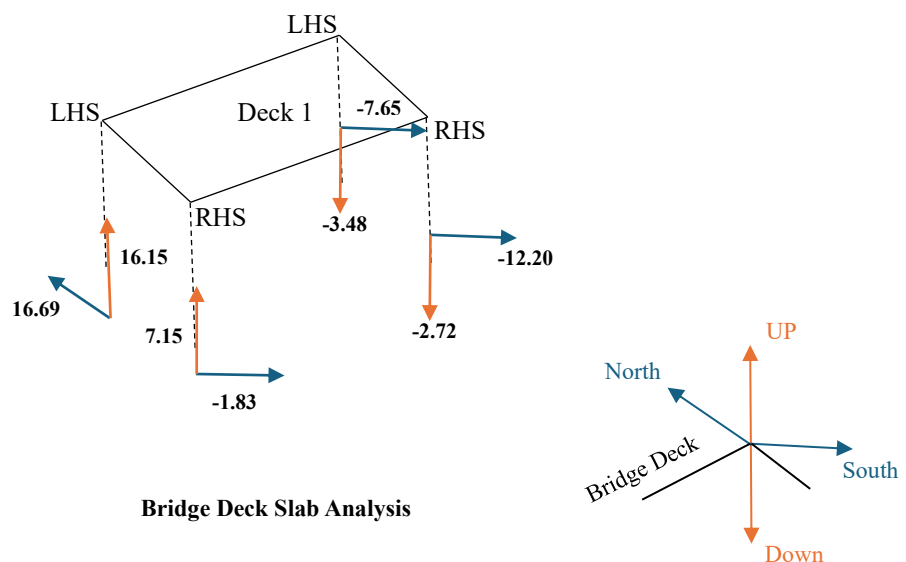
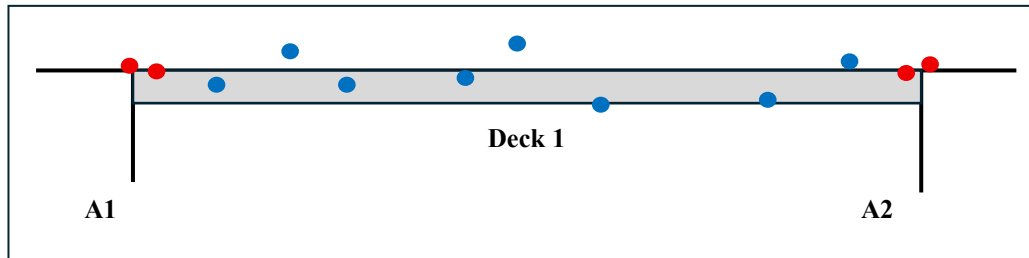
Critical Events & Anomalies:

- **February 2025:** VV spikes sharply to $+13$ mm uplift, while VH continues negative (-10 mm), revealing a decoupled vertical rebound without horizontal correction.
- **May 2025:** VH reaches its maximum negative magnitude, while VV oscillates near $+4$ mm, indicating severe lateral movement with reduced vertical stiffness.
- **June 2025:** VV drops further to -18 mm, while VH again deepens to -22 mm, marking the most critical end-of-period displacement state.

ANALYSIS

1. Identify potential deformation, shifts or movement of the bridge structure

- Points of Analysis (POA) of bridge (●)



Vertical Displacement Behaviour:

- The bridge deck slab exhibits strongly non-uniform vertical displacement, indicating pronounced differential vertical response between the LHS and RHS regions. The maximum vertical uplift is observed at the LHS support region, reaching approximately +16.15 mm, representing the dominant upward response of the deck.

Horizontal Displacement Behaviour:

- Horizontal displacement shows clear directional variability across the deck width, reflecting torsional deformation behaviour. The central deck region shows moderate horizontal displacement of approximately -7.65 mm, while the lower central region records minimal movement. The opposing directions and uneven magnitudes of horizontal displacement strongly indicate torsional rotation about a longitudinal axis, rather than rigid-body translation.

CONCLUSION:

- The bridge deck slab demonstrates a torsion-dominated deformation response, characterized by large asymmetric horizontal displacement combined with significant vertical uplift and localized settlement. Overall, the magnitude and asymmetry of the observed displacements indicate structurally significant behaviour, warranting continued monitoring and detailed assessment, particularly with respect to torsional effects and support performance.